

AWS LAMP & WordPress Deployment with Cloud Enhancements

Overview

This project began as a cloud migration proof-of-concept using Amazon EC2, Apache, MariaDB, PHP, and WordPress. It was later enhanced with Route 53, Load Balancer, Auto Scaling, RDS, and EFS concepts to improve resilience, scalability, and administration.

Initial Deployment

The first stage involved launching an Amazon Linux EC2 instance, configuring HTTP/SSH access, installing the LAMP stack, creating a database, and deploying WordPress.

Fault Tolerance Enhancement

Route 53 and a Load Balancer provide a stable access point and route traffic away from unhealthy instances. Auto Scaling supports automatic replacement if an instance fails.

Automatic Demand Handling

Auto Scaling and load balancing were used to support traffic changes. ApacheBench was used to generate test traffic and observe system behaviour under load.

Reduced Administration

RDS was introduced to reduce local database management, while EFS was used as shared storage for WordPress uploads across instances.

Security Note

The raw report contains lab credentials and setup screenshots, so this public package uses sanitized documentation and redacted screenshots instead.

Key Services Used

- EC2
- Apache HTTP Server
- MariaDB / RDS
- PHP
- WordPress
- Route 53
- Application Load Balancer
- Auto Scaling Group
- EFS
- ApacheBench